

Wumpus World Variant Lava Cells and the Heat Percept

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Problem Analysis and AI System Design

1. Problem Analysis

Goals and Constraints

The agent's task is to traverse a 5x5 partially observable grid to retrieve the gold and make it back to the starting point (1,1) without getting trapped. There are three hazards to die on entry: Pits, the Wumpus and the new hazard Lava cells. The agent starts with only one arrow and no information on the location of hazards. Additionally, hazard placement is fixed at the beginning of each episode, the agent can move only in four cardinal directions, the agent can only see the cell he is in, and hazard cells cannot be accessed diagonally or through the grid's boundary.

Environment Description

Property	Value
Observable	Partial — agent perceives current cell only
Deterministic	Yes — all actions produce fixed outcomes
Sequential	Yes — earlier decisions affect future states
Static	Yes — layout is fixed; Wumpus death is the only state change
Discrete	Yes — finite set of cells and actions
Single-agent	Yes

The agent may receive up to six percepts at each cell:

Percept	Meaning
Stench	Wumpus is in an orthogonally adjacent cell
Breeze	A Pit is adjacent
Heat (new)	A Lava cell is adjacent
Glitter	Gold is in the current cell
Bump	Agent attempted to move into a wall
Scream	Wumpus has been killed

The Twist: Lava Cells and the Heat Percept

Lava is contained in one or more of the cells and is lethal on contact. Lava is unique among Pits (signaled by Breeze) and the Wumpus (signaled by Stench) in that it produces a different percept (Heat) in each orthogonally adjacent cell. This necessitates that for each type of hazard the agent has to keep three separate inference chains in parallel, one for each type of hazard, while keeping the knowledge base tractable for propositional logic, and its complexity high. The agent needs to avoid mixing the signals of both Lava and Pits, as both are completely independent of each other, and the reasoning problem is also interesting because of that.

2. Justification of Approach

A propositional logic inference agent is the most appropriate choice. The environment is a deterministic, discrete and each percept contains provably correct information. The rules like "If no Heat is detected at a visited cell, then unvisited neighboring cells have no Lava" can be guaranteed. Each of the decisions is fully traceable to symbolic rules to satisfy the assessment's explainability requirement.

A reactive agent (for example, random walk) is not appropriate because it does not have the ability to store knowledge over time. It would keep on threatening cells already determined to be hazardous and would not be able to take advantage of safe inferences that the logic agent makes.

A model-free reinforcement learning agent also does not work: it needs many episodes to converge, cannot guarantee safety in one episode, and leaves no "reasoning chain" of explainable reasoning steps that is vital to the value of this domain.

3. System Representation

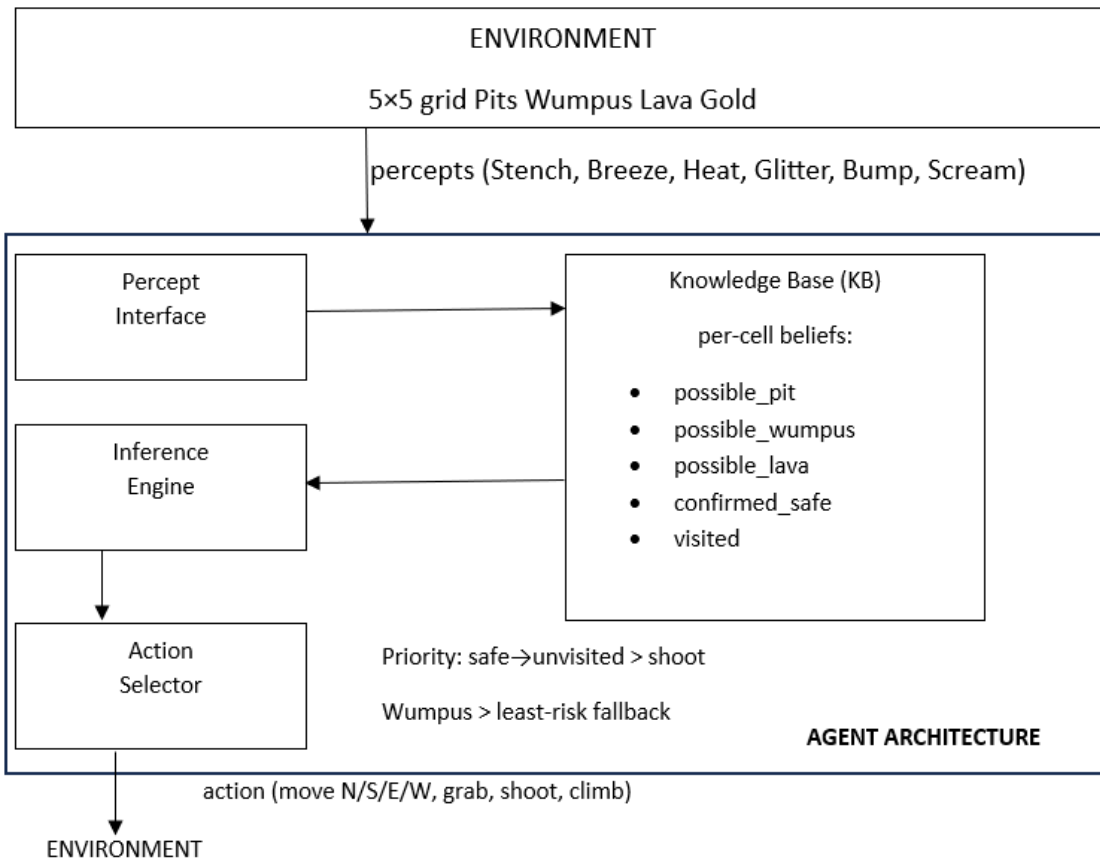


Figure 1: Agent architecture for the Lava Wumpus World variant.

4. Algorithmic Plan

INITIALISE:

```
KB ← empty knowledge base (all cells: unknown)
safe_cells ← {(1,1)}
visited ← {}
has_gold ← False
has_arrow ← True
```

EACH STEP:

```
percepts ← sense(current_cell)
add current_cell to visited
UPDATE_KB(current_cell, percepts)
INFER_SAFE_CELLS()
```

IF Glitter IN percepts:

```

    grab_gold()
    has_gold ← True

IF has_gold AND current_cell = (1,1):
    climb_out()
    STOP

IF unvisited safe cells exist:
    move_to(nearest unvisited safe cell via BFS)
ELSE IF wumpus_cell confirmed AND has_arrow:
    shoot_arrow(direction of wumpus_cell)
    has_arrow ← False
ELSE:
    move_to(unvisited cell with lowest combined hazard
probability)

UPDATE_KB(cell, percepts):
    FOR each orthogonal neighbour N of cell:
        IF Stench NOT in percepts: KB.no_wumpus(N) ← True
        IF Breeze NOT in percepts: KB.no_pit(N) ← True
        IF Heat NOT in percepts: KB.no_lava(N) ← True
        IF Stench in percepts: KB.poss_wumpus(N) ← True
        IF Breeze in percepts: KB.poss_pit(N) ← True
        IF Heat in percepts: KB.poss_lava(N) ← True

INFER_SAFE_CELLS():
    FOR each unvisited cell C:
        IF KB.no_pit(C) AND KB.no_wumpus(C) AND KB.no_lava(C):
            add C to safe_cells

```

Negative percepts are the most powerful signal: a single percept-free visit to a cell rules out hazards in all its neighbors simultaneously, consistent with standard propositional Wumpus logic (Russell and Norvig, 2021).

Evaluation, Reflection, and Ethical Analysis

1. Reflection on My System

The implemented agent derives conclusions from propositional logic about a 5x5 grid with three independent types of hazards: Pits, Wumpus and Lava cells. Each step, the agent modifies a symbolic knowledge base based on percepts and then performs an action based upon the rules of inference. The system is determinate, that is, the decisions made will be the same if the same sequence of percepts is presented on the same grid.

Strengths

The agent's reasoning is fully transparent. Each action is explained by a clearly-stated rule, such as "Heat was not present at (4,0) and (4,1), so there was no lava at (3,0) and (3,1). This allows errors to be quickly identified and rectified. What's more, the system cannot react in a way not expected by its designer: the system doesn't have the ability to find unexpected patterns or act beyond what the rules specify.

Limitations

The agent's inference is fragile, in ambiguous contexts. If more than one unvisited cell is able to account for the same percept, the system is not able to reason about which one is more likely to be dangerous; it simply defaults to a "lowest hazard score" heuristic. The system is also incapable of learning: If a new type of percept was added, the rules would have to be rewritten. Finally, the 5x5 grid with 3 hardcoded hazard types is a trivialization of any actual navigation situation, and therefore, it is not easy to extrapolate the results of the 5x5 grid to other situations.

2. Advanced AI Contrast: Deep Reinforcement Learning

A more modern approach would be to train a deep reinforcement learning (DRL) agent using thousands of episodes of the Wumpus World, such as a Deep Q-Network (DQN) or Proximal Policy Optimization (PPO). Instead of explicitly programming rules, this approach would learn a policy that associates observations with actions by trial and error with the help of a reward signal (for example, +1000 if a gold coin is picked up, -1000 if the player dies, -1 if the player takes a step).

Capabilities

A DRL agent will be able to generalize without the need to manually update rules for different grid sizes and configurations. It might find novel solutions, like the temporal pattern of Heat percepts over several cells and triangulate the location of lava, which the designer might not have thought of. It would also scale naturally: if a fourth hazard type is added, only the reward function needs to be updated and the machine retrained, but not the design of all the inference rules.

Transparency and explainability

The contrast is stark here. A neural network is not human-readable and its weights are defined by a DRL agent's policy. There is no simple logical answer to the question "why did the agent move left and not right?". Partial explanations can be generated using post-hoc explanation methods (for example, saliency maps, LIME), but these are only approximations of the network's actual reasoning, not the actual reasoning.

Unintended consequences

Also, known as reward hacking, DRL agents exploit reward functions which designers didn't foresee. You might find that an agent could get the best rewards out of the game by not going for the gold at all, because they will receive a penalty if they die. It may turn out that the agent could get the maximum possible reward from the game by never going for the gold because if they die, they get a penalty.

3. Ethical Reflection and Trade-offs

Transparency and Explainability

My rule-based system is designed to be fully explicable. This is important, not just in terms of assignments but also as a general rule: If an AI agent makes decisions that have consequences (for example, navigation, advice, diagnosis), then their ability to explain the reasons for the decision facilitates human oversight and trust. This is given up entirely by the DRL alternative. AI systems are increasingly being applied in more critical contexts, such as medical diagnosis, autonomous driving and judicial risk assessment, and the opacity of neural networks is no longer just a technical issue, but an ethical one. If an agent can't answer why it took a certain course, they can't be held accountable for it.

Accountability for Mistakes

In my system, if the player makes a mistake (let's say they go into an ambiguous cell, which turns out to be deadly), they will be able to refer back to a particular missing inference rule, or an unlucky placement of cells in the grid. The responsibility lies with the rule's designer. A DRL system has diffuse sources of error, data, reward function, architecture, and emergent

behavior, none of which is directly responsible for the source of error. This helps spread the responsibility and can make it possible to shift the blame forever.

Bias and Fairness

Unlike data-driven agents, Propositional Logic agents don't have a training set and thus no data-based bias. All rules are applied to every cell, no matter where it is located. If a DRL agent is trained on a certain distribution of grid configurations, then it might be prone to systematically misbehaving (Mnih et al. 2015), that is, a grid that has a lava cell in a certain position will be more likely to have a lava cell in the same position when tested. This is a minor point in the Wumpus domain, but is a major point when extrapolating to real-world navigation or decision making.

Control and Predictability

My agent is deterministic and its behavior is completely predictable based on its rules. A DRL agent, good on average, can be bad at the extremes of its training distribution. The differences between episodes; strategies they find, the failure modes that occur; are unpredictable. This unpredictability is an ethical issue, not just a technical constraint, in a domain in which predictability is essential for safety (industrial robotics, medical assistance).

Trust and Over-reliance

My system is very easy to use and because of its simple design and its known shortcomings, the user is not likely to take it beyond its capabilities. There is little risk of over-reliance. A DRL system that does a good job on a lot of test cases can give a false sense of robustness (Ribeiro et al. 2016). Users, including those creating software, can rely on it to exceed its limits of reliability, and suffer harm as a result.

Does Greater Capability Mean Better Ethics?

As the comparison above indicates, technical strength is not directly correlated with ethical results. The DRL system is capable of addressing more complicated problems, of generalizing more broadly and of needing less manual design, but by doing so, it becomes less explainable, less accountable, and less predictable. These do not need to be designed out of a learning paradigm; they are properties of a learning paradigm. Essentially, the ethical AI system needs to be able to perform its job well, while also being transparent, and ensuring that accountability and control are achievable even as its performance improves. Even though this is a partial, and flawed, model of what AI reasoning could be like, it helps to provide a benchmark, one that can be evaluated against more powerful models.

References

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